Multiple access comparison overview

SIMATS / MOBILE COMPUTING

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Comparison of Multiple Access Techniques

FDMA, TDMA, CDMA and OFDMA in Mobile Communication

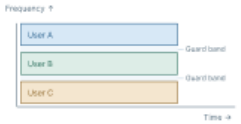
Overview | Techniques | Comparison Table | Summary

User A User B User C

Multiple access techniques allow multiple users to share a common communication channel by separating their transmissions using frequency, time, codes, or orthogonal subcarriers.

FDMA Frequency Division Multiple Access

Detail →



Concept: Each user is assigned a separate frequency channel.

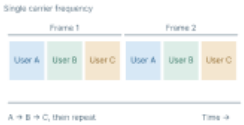
Advantages: Simple separation; simultaneous transmission on different frequencies.

Limitations: Guard bands; inefficient allocation when traffic varies.

Example: Traditional analog cellular systems.

TDMA Time Division Multiple Access

Detail →



Concept: Users share the same carrier frequency but transmit in different time slots.

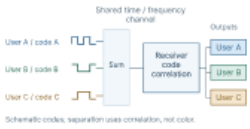
Advantages: Organized sharing; multiple users share one carrier.

Limitations: Accurate synchronization; users wait for assigned slots.

Example: GSM: FDMA + TDMA.

CDMA Code Division Multiple Access

Detail →



Concept: Multiple users share the same frequency band and time interval using different spreading codes.

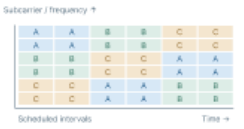
Advantages: Shared spectrum; code-based resource sharing.

Limitations: Code management, power control and interference.

Example: IS-95 and CDMA2000.

OFDMA Orthogonal Frequency Division Multiple Access

Detail →



Concept: Users receive different groups of orthogonal subcarriers across scheduled time intervals.

Advantages: Flexible, efficient allocation for different traffic needs.

Limitations: Frequency offsets, synchronization and high PAPR.

Example: LTE downlink and 5G NR.

Comparison Table

Simultaneous access is compared within the stated shared resource.

Technique	Resource Separation	Simultaneous User Access	Main Advantage	Main Limitation	Example Application
FDMA	Separate frequency channels	Yes — different frequencies	Simple channel separation	Guard bands; variable-traffic inefficiency	Traditional analog cellular systems
TDMA	Different time slots	No — users alternate on one carrier	Organized carrier sharing	Synchronization; waiting for slots	GSM (FDMA + TDMA)
CDMA	Different spreading codes	Yes — same band and time, different codes	Shared-spectrum access	Code management; power control; interference	IS-95; CDMA2000
OFDMA	Orthogonal subcarriers + time	Yes — distinct orthogonal subcarriers per scheduled interval	Flexible, efficient allocation	Synchronization; frequency offsets; high PAPR	LTE downlink; 5G NR

Remember the Difference

FDMA = Different frequencies

TDMA = Different time slots

CDMA = Different spreading codes

OFDMA = Different groups of orthogonal subcarriers and time-frequency resources

FDMA technique detail

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Comparison of Multiple Access Techniques

FDMA

Frequency separation

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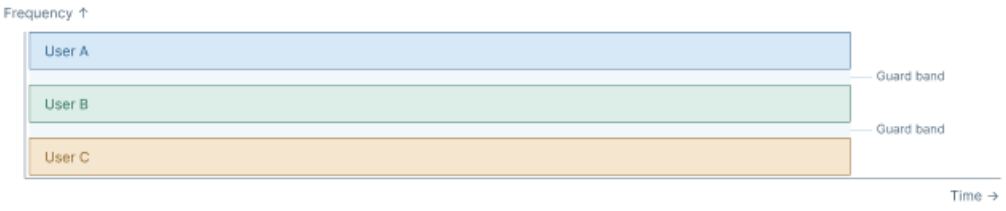
Frequency Division Multiple Access

CONCEPT

Each user is assigned a separate frequency channel.

Frequency channels over time

User A User B User C



Each horizontal band is a dedicated user channel. Guard bands between channels reduce adjacent-channel interference.

Advantages

- Simple channel separation.
- Users can transmit simultaneously on different frequencies.

Limitations

- Requires guard bands.
- Frequency allocation can be inefficient when traffic varies.

EXAMPLE APPLICATION

Traditional analog cellular systems.

TDMA technique detail

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Comparison of Multiple Access Techniques

TDMA

Time separation

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Time Division Multiple Access

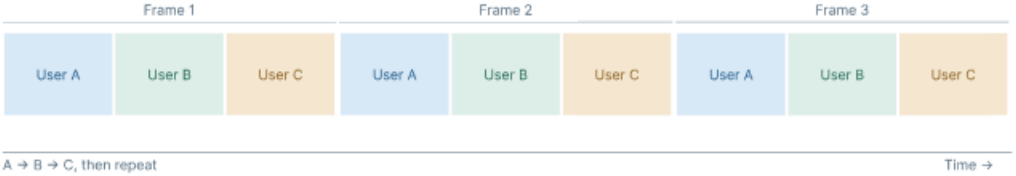
CONCEPT

Users share the same carrier frequency but transmit in different time slots.

Repeating time-slot allocation

User A | User B | User C

Single carrier frequency



One carrier is divided into repeating frames. Within each frame, users take turns in their assigned time slots.

Advantages

- Organized resource sharing.
- Multiple users can share a carrier.

Limitations

- Requires accurate time synchronization.
- Users may wait for their assigned slots.

EXAMPLE APPLICATION

GSM uses a combination of FDMA and TDMA.

CDMA technique detail

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Comparison of Multiple Access Techniques

CDMA

Code separation

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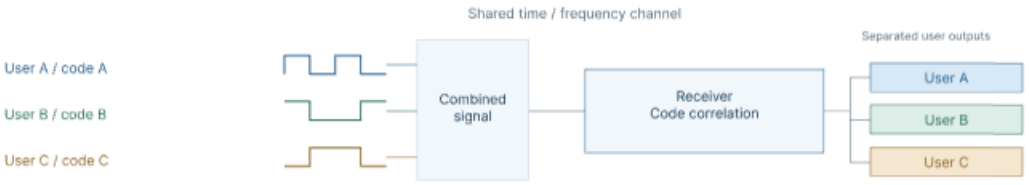
Code Division Multiple Access

CONCEPT

Multiple users share the same frequency band and time interval using different spreading codes.

Shared channel and code-based recovery

User A | User B | User C



The receiver correlates the combined signal with each user's spreading code. The waveforms are schematic code patterns, not separation by color.

Advantages

- Supports shared-spectrum access.
- Code-based separation allows multiple users to share resources.

Limitations

- Requires code management and power control.
- Interference can affect performance.

EXAMPLE APPLICATION

IS-95 and CDMA2000.

OFDMA technique detail

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Comparison of Multiple Access Techniques

OFDMA

Time–frequency allocation

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Orthogonal Frequency Division Multiple Access

CONCEPT

Users receive different groups of orthogonal subcarriers across scheduled time intervals.

Scheduled time–frequency resource grid

User A | User B | User C

Subcarrier / frequency ↑

A	A	B	B	C	C
A	A	B	B	C	C
B	B	C	C	A	A
B	B	C	C	A	A
C	C	A	A	B	B
C	C	A	A	B	B

Scheduled time intervals • allocation changes with demand

Time →

At each scheduled interval, different users occupy different orthogonal subcarriers. Assignments can change with time and traffic demand.

Advantages

- Flexible resource allocation.
- Efficient spectrum use.
- Supports different traffic requirements.

Limitations

- Sensitive to frequency offsets.
- Requires accurate synchronization.
- Can have high peak-to-average power ratio (PAPR).

EXAMPLE APPLICATION

LTE downlink and 5G NR.